Building Human-Centered AI Applications: Syllabus

IST 688 | Fall 2026

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Course Description

This hands-on course teaches techniques for developing and deploying intelligent applications powered by large language models. Students will gain technical skills in prompt engineering, knowledge enhancement, conversational interface design, and evaluating model outputs. Responsible AI development practices related to ethics, bias mitigation, and accessibility will be woven throughout. No prior AI experience required. Final projects provide an opportunity to apply concepts to build an AI system.

Class Meetings

Tuesdays, 9:30 AM – 12:15 PM in Hinds Hall 117

Office Hours

My regular office hours are **Thursdays, 1:00 PM – 2:00 PM in Hinds Hall 309**. There is no need to make an appointment for regular office hours; just drop in. I understand that this time may not work for everyone, so alternative times can be arranged. The easiest way to schedule an alternative meeting time is to email me a few days in advance of when you want to meet and include a list of days and times that you will be available. Note that office hours aren't just for problems with homework or feedback on a grade; sometimes you just want to chat or seek guidance about classes to take in a future semester. If you think you may need a letter of recommendation from me in the future, office hours are a good way for me to get to know you.

Never hesitate to drop into my office hours or reach out for an appointment!

Communications

I prioritize communication by email. Please do not use Teams or messages within Blackboard because I am not likely to see them in a timely manner. When communicating by email, the subject line should read "IST 688: Brief description of question/issue". Note that I generally do not correspond outside of standard working hours, including weekends, so please plan accordingly.

Audience

Graduate students

Course Prerequisites

Familiarity with the Python programming language

Credits

3 credit hours

Learning Objectives

After taking this course, students will be able to:

1. Explain concepts of large language models
2. Engineer effective prompts by applying techniques for tone, formatting, conditional logic, and mitigating risks
3. Develop applications leveraging large language model APIs and libraries
4. Implement memory and knowledge techniques such as retrieval augmentation to enhance application capabilities
5. Describe how applications can integrate LLMs (Large Language Models) with actions (such as sending an email) and external data stores (databases, data services)
6. Build ethical AI systems by considering potential biases, harms, and real-world impacts

Textbook

There is no required textbook. However, the following resources will be helpful:

- OpenAI: https://platform.openai.com/docs/overview
- OpenAI Cookbook: https://github.com/openai/openai-python
- Claude: https://platform.claude.com/docs/
- Streamlit: https://docs.streamlit.io/get-started

Tools Used in this Course & Associated Costs

This course uses free, open-source software such as Python and Streamlit. Some assignments and the final project require paid APIs from providers such as OpenAI and Anthropic. Students are responsible for API costs, typically estimated at $10–$30 for the semester, though expenses may vary and could be higher. Students concerned about these expenses should contact the instructor early to discuss options.

Grading

Your grade in this course will be determined using the following assessments.

- **Lab assignments**: 9 × 1% = **9%** — Done in class; usually due Thursday nights (see Blackboard for details); no late submissions.
- **Homework assignments**: 7 × (2–4%) = **20%** — Due Sunday night, the week they are assigned; late submissions discounted 25% per day late.
- **Exam**: **25%** — Paper-based; no book/notes.
- **Team topic presentation**: **15%** — Half presentation, half lab assignment preparation; topics are assigned and structured.
- **Team project**: **25%** — Groups of 2-3; details provided later in the semester.
- **Class participation**: **6%** — Be an active participant in class (active listening and offering thoughts as appropriate); random attendance may be taken, especially during student-led presentations.

All work will be graded on the scale as specified for the component. All individual grades are summed at the end of the semester, and the final letter grade is determined using the following scale. These intervals are firm — I will not round grades up to qualify for the next letter grade cutoff.

A	A−	B+	B	B−	C+	C	C−	F
100–93	92–90	89–87	86–83	82–80	79–77	76–73	72–70	69–0

Course Specific Policies

- It is unethical to allow some students additional opportunities, such as extra credit assignments, without allowing the same options to all students
- Lab assignments are due Thursdays; no late submissions will be accepted for credit
- Homework assignments are due Sundays; late submissions are discounted 25% per day late

Use of Artificial Intelligence

Based on the specific learning outcomes and assignments in this course, artificial intelligence is permitted on take-home assignments and the final project. See each assignment's instructions for more information about what artificial intelligence tools are permitted and to what extent, as well as citation requirements. If no instructions are provided for a specific assignment, then no use of any artificial intelligence tool is permitted. Any AI use beyond that which is detailed in course assignments is explicitly prohibited except when documented permission is granted.

Course Schedule

The following schedule is subject to change.

Reminder: Lab assignments are due Thursdays and homework assignments are due Sundays.

Week #: Date	Topics	Assignments Due
01: Aug 25	Introduction to Large Language Models (LLMs); Using the OpenAI API and Streamlit	Lab 1; HW 1
02: Sep 01	Basic applications; More Streamlit; More LLMs	Lab 2; HW 2
03: Sep 08	Basic conversational agents & short-term memory; Prompt engineering	Lab 3; HW 3
04: Sep 15	Retrieval-Augmented Generation (RAG)	Lab 4; HW 4
05: Sep 22	Using tools/functions with LLMs; Overview of group presentations	Lab 5; HW 5; Select groups
06: Sep 29	AI in the real world / Group HW assignment	HW 6
07: Oct 06	Present HW 6; Responsible & ethical AI	—
—	**Fall Break**	HW 7
08: Oct 20	**Exam**	—
09: Oct 27	*Student Presentations*: (1) OpenAI Responses API (2) LangChain	Lab 6
10: Nov 03	*Student Presentations*: (1) Running models locally (2) Model fine-tuning	Lab 7; Project proposal
11: Nov 10	*Student Presentations*: (1) Audio/images (2) RAG and ReRanking	Lab 8; Project approach & milestones
12: Nov 17	*Student Presentations*: (1) Long-term memory (2) Multi-agent systems	Lab 9; Project review
—	**Thanksgiving Break**	—
13: Dec 01	Project work	—
14: Dec 08	Project presentations	Final project